Task	Prompt	Answer
neighbor	The task is to determine the neighbors of a node in the graph. For directed graph, you should return the successors of the node. Here is an undirected graph containing nodes from 1 to 10. The edges are: (1, 3), (1, 9), (1, 6), (1, 7), (3, 2), (3, 8), (3, 9), (6, 7), (2, 10), (10, 8), (4, 5). Question: What are the neighbors of node 2 in the graph? You need to format your answer as a list of nodes in ascending order, e.g., [node-1, node-2, ..., node-n].	[3, 10]
node number	The task is to determine the number of nodes in the graph. Here is an undirected graph containing nodes from 1 to 10. The edges are: (1, 10), (1, 3), (10, 6), (10, 8), (3, 7), (3, 4), (2, 7), (2, 5), (2, 9), (5, 9), (5, 8), (9, 4), (8, 6). Question: How many nodes are there in the graph? Your answer should be an integer.	10
periphery	The task is to determine the periphery of a graph. The periphery of a graph is the set of nodes with the maximum eccentricity. The eccentricity of a node is the maximum distance from this node to all other nodes in the graph. The input graph is guaranteed to be connected. Here is an undirected graph containing nodes from 1 to 6. The edges are: (1, 3), (3, 2), (3, 4), (3, 5), (3, 6). Question: What is the periphery of the graph? You need to format your answer as a list of nodes in ascending order, e.g., [node-1, node-2, ..., node-n].	[1, 2, 4, 5, 6]
radius	The task is to determine the radius of a graph. The radius of a graph is the minimum eccentricity of any node in the graph. The eccentricity of a node is the maximum distance from this node to all other nodes in the graph. The input graph is guaranteed to be connected. Here is an undirected graph containing nodes from 1 to 5. The edges are: (1, 2), (2, 3), (3, 4), (3, 5), (4, 5). Question: What is the radius of the graph? You need to format your answer as a float number.	2
resource allocation index	The task is to determine the resource allocation index of two nodes in a graph. The resource allocation index of two nodes is the sum of the inverse of the degrees of the common neighbors of the two nodes. The input graph is guaranteed to be undirected. Here is an undirected graph containing nodes from 1 to 5. The edges are: (1, 2), (1, 3), (2, 3), (3, 4), (3, 5), (4, 5). Question: What is the resource allocation index between node 1 and node 4? You need to format your answer as a float number.	0.25
shortest path	The task is to determine the shortest path between two nodes. The input nodes are guaranteed to be connected. Here is an undirected graph containing nodes from 1 to 6. The edges are: (1, 2), (1, 3), (2, 4), (2, 3), (2, 5), (3, 4), (3, 5), (4, 6). Question: What is the shortest path between node 1 and node 6? You need to format your answer as a list of nodes, e.g., [node-1, node-2, ..., node-n].	[1, 2, 4, 6]